

INFORMATION

Title: Design C — Martini Glass | **Sheet No:** Sheet #4

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Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels [P1-P4], each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1-F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3: **dot strip plot** (investment dispersion across LGAs, Q3), **bump chart** (mode share rank shift across investment tiers, Q1), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT

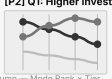
[A] State A — Guided Scroll

[P1] Q3: Investment broadly follows households



Dot Strip — LGA Investment

[P2] Q1: Higher investment LGAs show more car use



Bump — Mode Rank × Tier

[P3] Q2: Wealthier neighbourhoods own more vehicles



Connected Dot × CI

[P4] Now explore the evidence yourself

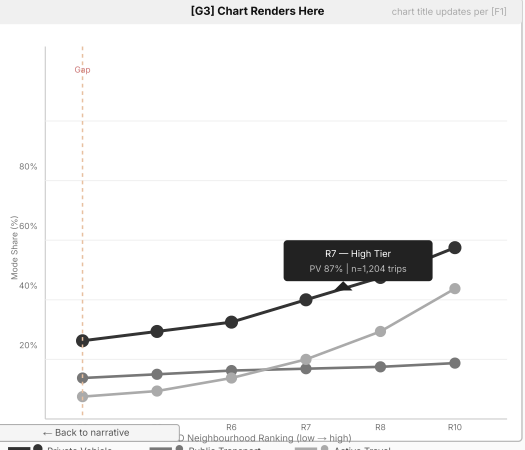
[D] Jump to Explore →

[H] Source ↗

[B] State B — Open Exploration

[G3] Chart Renders Here

chart title updates per [F1]



← Back to narrative

Neighbourhood Ranking (low → high)

● Private Vehicle ● Public Transport ● Active Travel

Filters Active

[F1] Metric

Mode Share

→ changes chart type in [G3]

[F2] Trip Purpose

All (Work + Edu)

→ re-renders [G3]

[F3] Investment Tier

None

Low

Medium

High

→ compares with other filters

[F4] IRSD Range

Decile 2 — 10

→ removes/greys points outside range

Hover on [G3] →

value, group label, ranking, tier, n trips

⚠ VISTA Coverage Gap

29 LGAs: investment recorded, no VISTA data

Shown as grey stripes in [G3] when relevant

OPERATIONS

[A] Scroll / natural read through [P1-P3]

⇒

Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[F1] Change Metric dropdown

⇒

[G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2-F4] remain applied.

[F2] Change Trip Purpose dropdown

⇒

[G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes

⇒

[G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider

⇒

Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PARTI — TWO-STATE TRANSITION

State A — Guided Scroll

[P1] Dot Strip — Investment/LGA

[P2] Bump — Mode Rank × Tier

[P3] Connected Dot+CI — Vehicle Own.

[P4] Bridge: "Explore the data yourself"

Click [D]

one-way

Filters unlock from grey state

Back always available

State B — Open Exploration

[F1] Metric → changes chart type in [G3]

[F2] Trip Purpose → re-renders [G3]

[F3] Invest. Tier → filter [G3] points

[F4] IRSD Range → grey/remove points

Filters have separate drop boxes

Hover [G3] → tooltip always on

Parti Note

Narrative first, exploration second — author decides what evidence matters, reader decides depth.

3 chart types, none repeated from Sheets 2/3

[F1] switches type in [G3]

Chart Type per [F1] Metric

Mode Share → Connected dot plot

Trip Duration → Violin + box plot

Vehicle Ownership → Connected dot + 95% CI

Label Reference

[A] State A Guided Scroll

[B] State B Open Exploration

[D] Jump to Explore button

[F1] Metric dropdown

[F2] Trip Purpose dropdown

[F3] Investment Tier checkboxes

[F4] IRSD Range slider

[G3] Chart Render Zone

[H] Data Source link

[P1-P4] Narrative panels

State A enforces the narrative arc: investment geography (Q3) → mode share consequence (Q1) → socioeconomic mechanism (Q2). State B hands control to a reader who has already absorbed the argument. The [D] transition is one-way by default; "← Back to narrative" restores State A if the reader needs context again. Chart types across all three panels are entirely distinct from Sheets 2 and 3 — dot strip plot, bump chart, and connected dot plot with 95% CI are used here for the first time in the FDS sequence.

DISCUSSION

Pros

- Policymakers receive a structured evidence briefing before controls are handed over — they understand the argument before they can challenge it.
- Small, paired chart-text accumulation in [P1-P3] is cognitively natural: one claim, one chart, one reading at a time.
- [D] gives immediate access to experienced policymakers wanting to explore the data without sitting through the full narrative.
- [P4] makes it clear to readers that they have finished reading and are entering a different mode — reducing the ambiguity of the transition.
- "← Back to narrative" link enables users to come back to the guided mode without losing exploration state.
- Three chart types in [P1-P3] are analytically distinct and none are repeated from Sheets 2 or 3 — guarantees genuine FDS design space coverage.

Cons

- For experienced policymakers, the static narrative guided mode may feel slow — they may jump to [D] immediately, bypassing the intended framing.
- Placing the exploration zone on the right side in State B may make the reader think there is an exploration mode before even reading the guided version.
- Four stacked narrative panels make the left column long — readers on small screens face significant scrolling before reaching the [D] bridge panel and exploration mode.
- [F1] switching the chart type in [G3] changes the visual encoding significantly — a reader who has not read the panel descriptions may be confused by the sudden appearance of a violin plot rather than a dot plot.